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Access the code, data, and analysis at https://github.com/jossl95/co-pub_networks

Intersectional Inequalities in Co-Authorship Networks: Gender, Origin, and Collaboration in Dutch Social Science

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1 Introduction

Social inequality in academia has been documented across a wide range of contexts and indicators. Gender disparities in academic careers are described in an extensive body of literature (e.g., [Barrios et al., 2013](#); [Bol, 2023](#); [Greska, 2023](#); [L. Kim et al., 2022](#); [Lutter & Schröder, 2016](#)), which has shown that women have lower salaries ([C. Kim & Shin, 2024](#)), obtain fewer grants ([Stadmark et al., 2020](#)), receive less impact for work that is often more innovative ([Hofstra et al., 2020](#)), and have shorter and less prolific careers than men ([Mulders et al., 2024](#); [Tolsma et al., 2025](#)). Moreover, the topics women disproportionately work on tend to be devalued in scientific careers, yielding lower prestige and weaker career returns ([C. Kim & Shin, 2024](#)). Beyond gender, scholars from underrepresented national and ethnic groups are less likely to obtain senior positions, have lower average publication and citation counts, and face slower career progression than majority-group men ([Hofstra et al., 2020](#); [Kozłowski et al., 2022](#); [Whittington, 2018](#)). Even when publication rates are comparable, women and minoritised scholars often receive fewer citations and less recognition, illustrating classic Matthew effects in contemporary academic careers ([Hofstra et al., 2020](#); [Li et al., 2022](#); [Merton, 1973](#)).

These unequal outcomes are closely tied to inequalities in opportunities and collaborations. Science is an inherently social enterprise: colleagues, departments, conferences, publishing venues,

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and committees are organised so that the social relationships surrounding an academic are central to their success (Burt, 1998; Stark et al., 2020). Academic social capital matters for access to the professoriate, which depends heavily on one’s mentors (Hofstra et al., 2022); for dissertation committees, which exhibit gender-homophilous tendencies in awarding cum laude PhD degrees (Bol, 2023); and for the size and composition of co-authorship teams, which is linked to whether contributions are disruptive or consolidating in the scientific landscape (Wu et al., 2019). Inequalities in outcomes therefore reflect, in part, unequal access to collaboration opportunities. Women tend to have fewer international collaborations and less access to prestigious, high-impact collaborations than men (Barrios et al., 2013; Kwiek & Roszka, 2021; Whittington et al., 2024). Scholars from countries outside the traditional scientific core similarly face barriers to entering central collaboration circuits and forming ties with highly visible partners (Hâncean et al., 2014; Khor & Yu, 2016; J. Kim et al., 2016). These differences in opportunity structures shape whose work becomes visible and whose careers advance.

The problem of academic gender inequality is frequently framed in terms of “leaky pipelines” (Greska, 2023) and “chilly climates” (Cabay et al., 2018) for women. These metaphors highlight that socialisation into science, and the path dependencies it creates in careers, can be fragile for women scientists. However, they are less specific about the concrete relational mechanisms through which exclusion occurs. From a social network perspective, the question becomes which network processes and structural positions restrict women’s access to promising career trajectories and supportive professional environments. Given the centrality of social relationships to academic success, collaboration networks provide a natural empirical setting in which to investigate these mechanisms.

Co-authorship is particularly important in this respect, as these shape productivity, visibility, and citation impact, and they evolve systematically over time (Abbasi et al., 2011; Ductor, 2015; Li et al., 2022; Moody, 2004; Newman, 2001; Wallace et al., 2012). As fields mature, co-authorship networks densify, cluster, and develop pronounced core–periphery structures, with a small number of highly connected scholars occupying central positions and many others remaining peripheral (Abbasi et al., 2011; Ferligoj et al., 2015; Hou et al., 2007). Network mechanisms such as homophily, triadic closure, preferential attachment, and brokerage organise collaboration opportunities and inequalities (Barabasi & Albert, 1999; Bojanowski & Czerniawska, 2020; Ebadi & Schiffauerova, 2015; Gondal, 2011, 2018). Preferential attachment and cumulative advantage imply that already well-connected, high-status scholars attract more collaborators and enjoy disproportionate returns (Ductor, 2015; Li et al., 2022), while homophily and triadic closure channel new ties along existing social and institutional boundaries. In practice, these mechanisms often reproduce gendered and origin-based segregation in collaboration networks (Gao et al., 2022; Kwiek & Roszka, 2021; Whittington et al., 2024). Co-authorship thus sits at the core of what science is about - discovering and publishing -

and the question of who gains access to well-publishing, reputable co-authors is a key component of the academic inequality puzzle.

Within this context, collaboration networks may perpetuate structural inequalities in several ways. First, there are gender-based differences in collaboration patterns and positions. Studies across disciplines show that women are less likely to occupy structurally powerful positions in co-authorship networks - such as high-degree hubs, brokers bridging structural holes, or authors spanning institutions and subfields - even when they have similar publication counts to men (Gao et al., 2022; Whittington, 2018; Whittington et al., 2024). Women’s collaborations are more often local or national rather than international (Kwiek & Roszka, 2021), and they are less frequently positioned as lead or senior authors in multi-authored work (Barrios et al., 2013). These patterns limit access to novel ideas, resources, and prestigious venues, which in turn constrains productivity and citation impact (Collischon & Eberl, 2021; Ductor, 2015). Moreover, gender differences emerge not only in where actors are located in the network, but also in the returns to those positions: comparable centrality or brokerage scores often yield lower pay, recognition, or promotion odds for women than for men (Contreras et al., 2025; Lalanne & Seabright, 2022; Whittington, 2018; Woehler et al., 2021). Collaboration networks therefore do not merely reflect existing gender hierarchies; they also magnify them by converting similar structural capital into unequal rewards.

Second, there are origin-based differences in collaboration networks. Origin- and nationality-based inequalities arise in who participates in international collaboration and who occupies central positions in global networks. Scholars from underrepresented nationality groups within core countries, are more likely to occupy peripheral positions (Hâncean et al., 2014; Khor & Yu, 2016; J. Kim et al., 2016; Kwiek & Roszka, 2021). These positional disadvantages reduce exposure to high-impact journals and influential collaborators, contributing to lower citation impact and slower career progression. Origin-based gaps are further reinforced by institutional prestige hierarchies (Allison & Long, 1987, 1990) and by database coverage biases: work from elite universities and from English-speaking or Western countries is more likely to be indexed, cited, and recognised (Khor & Yu, 2016; Li et al., 2022), while research from less visible institutions or non-core regions is underrepresented in major bibliometric databases and collaboration maps (Céspedes et al., 2025; Zheng et al., 2025). Collaboration networks thus participate in the reproduction of global epistemic inequalities, privileging certain national and institutional origins over others.

Gender- and origin-based differences in collaboration do not operate independently. Intersectionality provides a framework for understanding how these axes of inequality intersect. Intersectionality highlights that gender and origin (or race, ethnicity, nationality) are not independent dimensions, but mutually constitutive axes of social stratification (Crenshaw, 1989; King, 1988). From this perspective, gender penalties may intensify for international women, while shared nationality may mitigate gender gaps in forming ties (Kwiek & Roszka, 2021; van der Wal et al., 2021). Empirical work on science and higher education suggests that women are not uniformly disadvantaged; penal-

ties tend to be strongest for women from underrepresented nationality or ethnic groups, whereas majority-group women sometimes experience more buffered disadvantages. Studies of international collaboration show, for instance, that women from less prestigious institutions or from peripheral systems are particularly unlikely to access high-status international partnerships (Kwiek & Roszka, 2021).

Network status processes may amplify these intersectional patterns. Processes related to visibility, centrality, gatekeeping, and sponsorship can compound for women from underrepresented nationality groups, while shared nationality may buffer gender penalties in some contexts (L. Kim et al., 2022; Whittington, 2018; Whittington et al., 2024). Recent work on co-authorship networks indicates that women are systematically less likely to occupy high-status bridging positions that connect institutions and subfields (Whittington et al., 2024), and that topic choices and “gendered knowledge” interact with gender and race in shaping who gains visibility and career rewards (Hofstra et al., 2020; L. Kim et al., 2022). Large-scale intersectional bibliometric analyses further show that women and scholars of colour (or non-dominant nationalities) are concentrated in less prestigious topics and receive fewer citations than majority-group men, even when they contribute equally novel or impactful work (Hofstra et al., 2020; Kozłowski et al., 2022). Approaches that simply include main effects of gender and origin therefore understate the disadvantage that accrues at their intersection and risk missing the specific penalties experienced by intersectional groups (Kozłowski et al., 2022).

This paper makes two main contributions to these debates. First, we study gender- and origin-based inequalities in collaboration using comprehensive data on actors, institutions, and publications. We analyse a near-population of social scientists, their institutional affiliations, and their publication and co-authorship histories. This allows us to characterise disparities in collaboration opportunities and outcomes not just for a selected discipline or journal, but for the entire social-science system under study, paralleling recent full-population approaches in the science-of-science literature (Hofstra et al., 2020; Whittington et al., 2024). We explicitly model path dependencies - endogenous network processes such as homophily, assortativity, activity, popularity, and triadic closure - using longitudinal network models. Whereas many studies interpret gender or origin gaps in collaboration without representing these endogenous processes (Gao et al., 2022; e.g., Kwiek & Roszka, 2021), we examine how much of the observed inequality remains once structural mechanisms are taken into account. This shifts the focus from descriptive differences to the network processes through which inequalities are reproduced. Because our data provide rich information on all actors in the network and approximate a full-population sample, we can also address survivor bias and sample selection problems that arise in bibliometric studies focused only on highly productive scholars or selected journals (Hâncean et al., 2014; Ossenblok et al., 2014). We therefore assess inequalities in collaboration not only among “survivors” at the top, but across the full spectrum of social-science careers.

Second, we introduce an explicitly intersectional perspective to the study of scientific collaboration. Rather than analysing gender and origin inequalities separately, we model their joint effects. Building on intersectional work in the science of science (Hofstra et al., 2020; Kozłowski et al., 2022), we treat gender $\times$ origin as a joint axis of stratification and ask whether collaboration patterns and network returns differ systematically across intersectional groups (e.g., native men, native women, international men, international women). We assess whether penalties associated with gender and origin are additive (each dimension contributes separately), compounding (penalties multiply for, say, international women), or buffering (shared nationality reduces gender gaps, or shared gender reduces origin gaps). We also investigate how key network processes - homophily, assortativity, triadic closure, and status-bridging - operate differently across intersectional groups. Specifically, we ask whether homophily on gender and origin is stronger for majority-group men than for minoritised scholars, whether international women are particularly excluded from triadic closure processes, and whether bridging ties to high-status alters are unequally distributed (Kwiek & Roszka, 2021; Whittington et al., 2024). By focusing on these mechanisms, we identify where in the collaboration process intersectional penalties emerge.

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- What are the research questions?
 - **RQ1:** To what extent are there gender- and origin-based differences in collaboration (networks)?
 - * At the ego level, we compare differences in the number of collaborations (degree), brokerage and bridging (betweenness, structural holes/constraint), and local clustering across gender and origin groups. This follows prior work that links network position to productivity and impact (Collischon & Eberl, 2021; Ductor, 2015; Li et al., 2022).
 - * At the network level, we examine global clustering, transitivity, assortativity by gender and origin, and the overall core-periphery structure of the co-authorship network, building on studies of national and disciplinary collaboration structures (Abbasi et al., 2011; Ferligoj et al., 2015; Hâncean et al., 2014; Hou et al., 2007).
 - **RQ2:** To what extent are there gender- and origin-based differences in homophily? Do actors tend to choose collaborators who share their gender or origin?
 - * At the dyadic level, we estimate homophily and status asymmetries using longitudinal tie-formation models (e.g. SAOMs or related dynamic network models) that include endogenous structural effects (closure, activity, popularity) and covariate effects for gender and nationality using ego, alter, and similarity terms (egoX, altX, sameX). This allows us to quantify whether, net of structural tendencies and productivity, scholars systematically prefer same-gender or same-origin collaborators and whether majority

scholars are more likely to be selected as alters than minority scholars (Block et al., 2019; Kwiek & Roszka, 2021).

- **RQ3:** To what extent does gender- and origin-based homophily intersect? Do actors tend to choose collaborators who share both their gender and their origin?
 - * We extend the dyadic models to include interaction terms for gender $\times$ origin similarity, and to differentiate homophily for intersectional groups. This enables us to test whether, for example, international women exhibit particularly strong same-origin homophily (perhaps as a strategy of protection), or whether majority-group men exhibit strong same-gender, same-origin closure that excludes others. By comparing these patterns across groups, we can assess whether intersectional homophily contributes to the reproduction of intersectional inequalities observed in network positions and outcomes (Hofstra et al., 2020; Kozlowski et al., 2022; Whittington et al., 2024).
- **RQ4:** Do key network processes - such as triadic closure and status-bridging - operate differently across gender, origin, and their intersection?
 - * Using longitudinal tie-formation models, we examine whether structural effects that capture triadic closure and status-bridging (i.e., forming ties to alters with high status, centrality, or h-index) vary by gender, origin, and gender $\times$ origin. Concretely, we allow the strength of closure and status-related popularity effects to differ between intersectional groups, in order to assess whether majority-group men are more likely to benefit from closure and high-status collaborations than women and minoritized scholars, and whether international women are systematically excluded from such status-bridging processes.

<description of empirical case goes here>

 Tip

I currently have chosen *Social Networks* as the target journal. Some points for discussion:

- do you feel that the current introduction does adhere to this journal's standards and conventions?
- this journal requires ethical approval for be eligible for publishing. Has this project required ethical approval?

 Note

Some points for consideration while reading:

- When writing the current draft I have not considered the length of the text. Admittingly, the text in it's current form is on the longer side. Directions on shorting the introduction is welcomed.
- I'm currently not happy with the term "origin", alternatives like ethnicity or nationality have been considered but do also not completely do justice to the operationalization
- For the arguments for origin-based differences in outcomes and (network) opportunities have been predominantly sourced from the status quo in the science of science literature. One argument that is currently lacking is the potential for "bias/discrimination", which has currently not been convincingly made in this literature. Do we want to explicitly state the potential for (implicit) discrimination or do we want to adhere to the traditions in this literature.
- I have intentionally made the intersectional argument as concise as possible, do you feel that the argument as such is sufficient?
- I have reformulated the research questions to be even more substantive. Admittingly, the explanatory research questions remain on the dyadic level. Do you have suggestions to also include a ego, network level in here?
- I have made the levels (ego, dyad, network) implicit in the argument thus far out of concern for the argument getting too extensive when incorporating these levels explicitly, do you feel that the current argument requires such explication?

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